

Seminar Nr. 1, Probability and Statistics

Euler's Functions; Counting, Outcomes, Events

Theory Review

Euler's Gamma Function: $\Gamma : (0, \infty) \rightarrow (0, \infty), \Gamma(a) = \int_0^{\infty} x^{a-1} e^{-x} dx.$

1. $\Gamma(1) = 1;$
2. $\Gamma(a+1) = a\Gamma(a), \forall a > 0;$
3. $\Gamma(n+1) = n!, \forall n \in \mathbb{N};$
4. $\Gamma\left(\frac{1}{2}\right) = \sqrt{2} \int_0^{\infty} e^{-\frac{t^2}{2}} dt = \int_{\mathbb{R}} e^{-t^2} dt = \sqrt{\pi}.$

Euler's Beta Function: $\beta : (0, \infty) \times (0, \infty) \rightarrow (0, \infty), \beta(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx.$

1. $\beta(a, 1) = \frac{1}{a}, \forall a > 0;$
2. $\beta(a, b) = \beta(b, a), \forall a, b > 0;$
3. $\beta(a, b) = \frac{a-1}{b} \beta(a-1, b+1), \forall a > 1, b > 0;$
4. $\beta(a, b) = \frac{b-1}{a+b-1} \beta(a, b-1) = \frac{a-1}{a+b-1} \beta(a-1, b), \forall a > 1, b > 1;$
5. $\beta(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}, \forall a > 0, b > 0.$

Arrangements: $A_n^k = \frac{n!}{(n-k)!};$

Permutations: $P_n = A_n^n = n!;$

Combinations: $C_n^k = \frac{A_n^k}{P_k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\dots(n-k+1)}{k!}.$

De Morgan's laws:

$$\overline{\bigcup_{i \in I} A_i} = \bigcap_{i \in I} \overline{A_i} \quad \text{and} \quad \overline{\bigcap_{i \in I} A_i} = \bigcup_{i \in I} \overline{A_i}.$$

1.

- a) How many passwords can be formed using 4 distinct letters from the word "COMPUTER"?
- b) How many of them do not contain any vowels?
- c) How many of them contain exactly one vowel?
- d) How many of them are such that all the vowels are never together?

2. A firm offers a choice of 10 free software packages to buyers of their new home computer. There are 25 packages available, five of which are computer games, and three of which are anti-virus programs.

- a) How many selections are possible?

- b)** How many selections are possible, if exactly three computer games are selected?
- c)** How many selections are possible, if exactly three computer games and exactly two anti-virus programs are selected?

3. A person buys n lottery tickets. For $i = \overline{1, n}$, let A_i denote the event: the i^{th} ticket is a winning one. Express the following events in terms of $A_1, \dots, A_n$.

- a)** A: all tickets are winning;
- b)** B: all tickets are losing;
- c)** C: at least one is winning;
- d)** D: exactly one is winning;
- e)** E: at least two are winning.

4. Three shooters aim at a target. For $i = \overline{1, 3}$, let A_i denote the event: the i^{th} shooter hits the target. Express the following events in terms of A_1, A_2 and A_3 .

- a)** A: the target is hit;
- b)** B: the target is not hit;
- c)** C: the target is hit exactly three times;
- d)** D: the target is hit exactly once;
- e)** E: the target is hit exactly twice.